Session 3: Variance and Standard Deviation

Measures of Dispersion, Manual Calculations & Python Implementation

Problem 1: Step-by-Step Manual Variance Calculation

Dataset: Daily step counts over 7 days = `[4500, 7000, 5000, 8000, 4000, 9000, 6000]`

Step 1: Calculate the Mean (μ)

`μ = (4500 + 7000 + 5000 + 8000 + 4000 + 9000 + 6000) / 7 = 43,500 / 7 = 6,214.29 steps`

Step 2 & 3: Calculate Deviations ($x_i - \mu$) and Squared Deviations $(x_i - \mu)^2$

Day (i)	Steps (x_i)	Mean (μ)	Deviation ($x_i - \mu$)	Squared Deviation $(x_i - \mu)^2$
1	4,500	6,214.29	-1,714.29	2,938,788.78
2	7,000	6,214.29	+785.71	617,340.20
3	5,000	6,214.29	-1,214.29	1,474,499.20
4	8,000	6,214.29	+1,785.71	3,188,750.20
5	4,000	6,214.29	-2,214.29	4,903,082.78
6	9,000	6,214.29	+2,785.71	7,760,180.20
7	6,000	6,214.29	-214.29	45,920.20
Sum of Squared Deviations $\sum(x_i - \mu)^2$				**20,928,561.56**

Step 4: Compute Variance (σ^2)

Population Variance: $\sigma^2 = \sum(x_i - \mu)^2 / N = 20,928,561.56 / 7 =$ **2,989,794.51 steps^2**

(Sample Variance s^2 using N−1 = 20,928,561.56 / 6 = 3,488,093.59 steps^2)

Problem 2: Python Function for Standard Deviation

Task: Implement `calculate_standard_deviation(scores)` returning standard deviation rounded to 2 decimal places.

```
import math def calculate_standard_deviation(scores): if not scores: return 0.00 mean = sum(scores) / len(scores) variance = sum((x - mean) ** 2 for x in scores) / len(scores) std_dev = math.sqrt(variance) return round(std_dev, 2) # Test case exam_scores = [85, 90, 78, 92, 88] print(f"Exam Scores: {exam_scores}") print(f"Standard Deviation: {calculate_standard_deviation(exam_scores)}")
```

Problem 3: Comparing Spending Consistency on Zomato

Data: Friend A = [200, 200, 200, 200, 200] | Friend B = [100, 300, 150, 400, 50]

Friend A Stats

- Mean (μ): ₹200.00
- Variance (σ^2): 0.00
- **Standard Dev (σ): ₹0.00**

Friend B Stats

- Mean (μ): ₹200.00
- Variance (σ^2): 15,000.00
- **Standard Dev (σ): ₹122.47**

CONSISTENCY INTERPRETATION

Friend A has perfectly consistent spending (σ = ₹0.00) because every daily expense is identical to the mean. In contrast, Friend B's daily spending fluctuates wildly from ₹50 to ₹400, resulting in a high standard deviation (σ = ₹122.47), indicating high volatility and low predictability.

Problem 4: Startup Salary Spread Analysis

Dataset: Monthly salaries of 6 employees = [28000, 29000, 27000, 60000, 26500, 27500]

```
import statistics salaries = [28000, 29000, 27000, 60000, 26500, 27500] mean_sal = statistics.mean(salaries) # ₹33,000.00 p_var = statistics.pvariance(salaries) # 146,250,000.00 p_stdev = statistics.pstdev(salaries) # ₹12,093.39 print(f"Mean: ₹{mean_sal:,.2f} | Variance: {p_var:,.2f} | Std Dev: ₹{p_stdev:,.2f}")
```

STARTUP SALARY SPREAD INTERPRETATION

The high standard deviation (**₹12,093.39** relative to a mean of ₹33,000) highlights **significant compensation inequality** across the startup team. Five employees earn a tightly clustered salary between ₹26,500 and ₹29,000, while a single outlier (₹60,000, likely a executive or lead founder) severely distorts the team average.